一. 1. 解: 15

$$\begin{cases} \dfrac{\partial^2 u}{\partial x^2}+\dfrac{\partial^2 u}{\partial y^2}=0, & 0<x<a,\ 0<y<a. \\ u|_{x=0}=0,\ u|_{x=a}=0, \\ u|_{y=0}=f(x),\ u|_{y=a}=g(x) \end{cases}$$ 5

2. 解:

$$\begin{cases} \dfrac{\partial u}{\partial t}=a^2\dfrac{\partial^2 u}{\partial x^2}, & 0<x<l,\ t>0 \\ \dfrac{\partial u}{\partial x}\Big|_{x=0}=-\dfrac{q}{k},\ \dfrac{\partial u}{\partial x}\Big|_{x=l}=0, \\ u|_{t=0}=f(x) \end{cases}$$ 5

3. 解:

$$\begin{cases} \dfrac{\partial^2 u}{\partial t^2}=a^2\dfrac{\partial^2 u}{\partial x^2}, & 0<x<2,\ t>0 \\ u|_{x=0}=0,\ u|_{x=2}=0, \\ u|_{t=0}=\begin{cases} hx, & 0\le x\le 1 \\ -hx+2h, & 1\le x\le 2 \end{cases},\ \dfrac{\partial u}{\partial t}\Big|_{t=0}=0. & t>0. \end{cases}$$ 5

二. 解: 设 $U(x,t)=V(x,t)+W(x)$.

则 $W(x)$ 满足 $\begin{cases} \dfrac{\partial^2 w}{\partial x^2}-1=0 \\ w|_{x=0}=0,\ w|_{x=1}=1 \end{cases}$

$\therefore W(x)=\frac{1}{2}x^2+\frac{1}{2}x$

$\therefore V(x,t)$ 满足

$$\begin{cases} \dfrac{\partial V}{\partial t}=\dfrac{\partial^2 V}{\partial x^2}, & 0<x<1,\ t>0 \\ V|_{x=0}=0,\ V|_{x=1}=0, \\ V|_{t=0}=x-\frac{1}{2}x^2-\frac{1}{2}x=\frac{1}{2}x-\frac{1}{2}x^2 \end{cases}$$

设 $V(x,t)=X(x)T(t)$

令 ~~设~~ $\dfrac{T'}{T}=\dfrac{X''}{X}=-\lambda.$

$\therefore \begin{cases} X''+\lambda X=0 \\ X(0)=X(1)=0 \end{cases}$ ① ~~和~~ $\{$ ~~$T'+\lambda T=0.$~~

当 $\lambda<0$ 或 $\lambda=0$ 时 $X(x)\equiv 0$. 舍.

当 $\lambda>0$ 时，设 $\lambda=\beta^2$. 则方程① 通解为 $X(x)=C_1\cos\beta x+C_2\sin\beta x$.

代入初始条件. $X(0)=C_1\cos\beta\cdot 0=C_1=0$. $X(1)=C_2\sin\beta=0$.

则 $\beta=n\pi.(n=1,2\cdots)$ $\therefore$ $T'+\lambda T=0$ 写成 $T'+n^2\pi^2T=0$

$\therefore T_n(t)=C_n e^{-n^2\pi^2t}$

$\therefore$ ~~$U_n(x$~~ $V_n(x,t)=D_n e^{-n^2\pi^2t}\sin n\pi x$ $\cdots$

$\therefore V(x,t)=\sum_{n=1}^{\infty}D_n e^{-n^2\pi^2t}\sin n\pi x$

代入 $V(x,0)=\frac{1}{2}x-\frac{1}{2}x^2=\sum_{n=1}^{\infty}D_n\sin n\pi x$

$\therefore D_n=2\int_0^1(\frac{1}{2}x-\frac{1}{2}x^2)\sin n\pi x\,dx=-\frac{2}{(n\pi)^3}(-1)^n+\frac{2}{(n\pi)^3}=\frac{2}{(n\pi)^3}[1-(-1)^n]$

$\therefore V(x,t)=\sum_{n=1}^{\infty}D_n e^{-n^2\pi^2t}\sin n\pi x=\sum_{n=1}^{\infty}\frac{2[1-(-1)^n]}{(n\pi)^3}e^{-n^2\pi^2t}\sin n\pi x,(n=1,2,3\cdots)$

三. 解: 15

特征方程 $(dx)^2-dx\,dt-2(dt)^2=0$

$(dx-2dt)(dx+dt)=0$

$\therefore$ 特征线 $\xi=x-2t$

$\eta=x+t$

$\therefore$ $\frac{\partial^2 u}{\partial\xi\partial\eta}=0$. 则 $u(x,t)=f_1(x-2t)+f_2(x+t)$

代入初始条件得 $f_1(x)+f_2(x)=\sin x$ ①

$-2f_1'(x)+f_2'(x)=0$ ②

将②式积分. 可得 $-2f_1(x)+f_2(x)=C$ ③

将①式与③式联立. 解得 $\begin{cases}f_1(x)=\frac{1}{3}\sin x-\frac{C}{3}\\ f_2(x)=\frac{2}{3}\sin x+\frac{C}{3}\end{cases}$

$\therefore u(x,t)=f_1(x-2t)+f_2(x+t)$

$=\frac{1}{3}\sin(x-2t)+\frac{2}{3}\sin(x+t)$.

四. 解：对 y 做拉普拉斯变换. 可得 $u(x,y) \xleftrightarrow{L} U(x,p)$，$1 \leftrightarrow \frac{1}{p}$，$1+y \leftrightarrow \frac{1}{p}+\frac{1}{p^2}$

15

则 $\frac{\partial U(x,p)}{\partial y} = pU(x,p) - u(x,0) = pU(x,p) - 1$

$\therefore \frac{\partial}{\partial x}\left(\frac{\partial u(x,p)}{\partial y}\right) = p\frac{du}{dx} = \frac{1}{p}$

$\therefore \frac{du}{dx} = \frac{1}{p^2}$ $\quad \therefore U(x,p) = \frac{1}{p^2}\cdot x + C$

代入条件 $U|_{x=0} = \frac{1}{p}+\frac{1}{p^2}$ 则 $C = \frac{1}{p}+\frac{1}{p^2}$

$\therefore U(x,p) = \frac{1}{p^2}x + \frac{1}{p} + \frac{1}{p^2}$

$\therefore$ 做反拉普拉斯变换 $u(x,y) = xy + 1 + y$，$(x>0, y>0)$

五. 解：设在 $y>-1$ 内有一点 $M_0(x_0,y_0,z_0)$ 则在 $y<-1$ 内. 对应点 $M_1(x_0,-2-y_0,z_0)$.

~~取 $y>-1$ 内一点~~ 设 $y>-1$ 内任一点 $M(x,y,z)$.

15

则有格林函数 $G(M,M_0) = \frac{1}{4\pi}\left(\frac{1}{r_{MM_0}} - \frac{1}{r_{MM_1}}\right)$

$$= \frac{1}{4\pi}\left(\frac{1}{[(x-x_0)^2+(y-y_0)^2+(z-z_0)^2]^{\frac{1}{2}}} - \frac{1}{[(x-x_0)^2+(y+2+y_0)^2+(z-z_0)^2]^{\frac{1}{2}}}\right)$$

$$\therefore \left.\frac{\partial G}{\partial n}\right|_{y=-1} = -\left.\frac{\partial G}{\partial y}\right|_{y=-1}$$

$$= \frac{-1}{4\pi}\cdot\left(-\frac{1}{2}\right)\left[\frac{2(y-y_0)}{[(x-x_0)^2+(y-y_0)^2+(z-z_0)^2]^{\frac{3}{2}}} - \frac{2(y+2+y_0)}{[(x-x_0)^2+(y+2+y_0)^2+(z-z_0)^2]^{\frac{3}{2}}}\right]\Bigg|_{y=-1}$$

$$= -\frac{1}{2\pi}\cdot\frac{y_0+1}{[(x-x_0)^2+(-1-y_0)^2+(z-z_0)^2]^{\frac{3}{2}}}$$

$$\therefore u(M) = \frac{1}{2\pi}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\frac{y_0+1}{[(x-x_0)^2+(-1-y_0)^2+(z-z_0)^2]^{\frac{3}{2}}}\cdot f(x,z)\,dx\,dz$$

六. 解: 将函数 $f(x)$ 展开成贝塞尔函数 $J_1(\alpha_i x)$ 级数形式.

15

得 $f(x)=\sum_{i=1}^{\infty} A_i J_1(\alpha_i x)$ ，而 $f(x)=x^3\ (0\le x\le 1)$

则 $A_i=\dfrac{2}{J_2^2(\alpha_i)}\displaystyle\int_0^1 x\cdot x^3 J_1(\alpha_i x)\,dx$

令 $\alpha_i x=t$. 则 $dx=\frac{1}{\alpha_i}dt$.

则积分 $\displaystyle\int_0^1 x^4 J_1(\alpha_i x)\,dx=\int_0^{\alpha_i}\frac{t^4}{\alpha_i^4}J_1(t)\cdot\frac{1}{\alpha_i}dt$

$$=\frac{1}{\alpha_i^5}\int_0^{\alpha_i} t^2\cdot d\left(t^2J_2(t)\right)$$

$$=\frac{1}{\alpha_i^5}\left(t^4J_2(t)\Big|_0^{\alpha_i}-\int_0^{\alpha_i}2t^3J_2(t)\,dt\right)$$

$$=\frac{J_2(\alpha_i)}{\alpha_i}-\frac{2}{\alpha_i^5}\int_0^{\alpha_i}d\left(t^3J_3(t)\right)$$

$$=\frac{J_2(\alpha_i)}{\alpha_i}-\frac{2}{\alpha_i^5}\,t^3J_3(t)\Big|_0^{\alpha_i}$$

$$=\frac{J_2(\alpha_i)}{\alpha_i}-\frac{2}{\alpha_i^2}J_3(\alpha_i)$$

$$\therefore A_m=\frac{2}{J_2^2(\alpha_i)}\cdot\left(\frac{J_2(\alpha_i)}{\alpha_i}-\frac{2}{\alpha_i^2}J_3(\alpha_i)\right)$$

$$=\frac{2}{\alpha_iJ_2(\alpha_i)}-\frac{4J_3(\alpha_i)}{\alpha_i^2J_2^2(\alpha_i)}$$

$$\therefore f(x)=\sum_{i=1}^{\infty}\left[\frac{2}{\alpha_iJ_2(\alpha_i)}-\frac{4J_3(\alpha_i)}{\alpha_i^2J_2^2(\alpha_i)}\right]J_1(\alpha_i x).$$

七. 解: 定解问题为

10

$$\begin{cases}\dfrac{\partial^2 u}{\partial\rho^2}+\dfrac{1}{\rho}\dfrac{\partial u}{\partial\rho}+\dfrac{1}{\rho^2}\dfrac{\partial^2 u}{\partial\theta^2}=0, & 0<\rho<a,\ 0<\theta<\pi.\\ u|_{\rho=a}=\theta(\pi-\theta), & 0<\theta<\pi,\ \rho=a.\\ u|_{\theta=\pi}=u|_{\theta=0}=0\end{cases}$$

$\therefore$ 设 $u(\rho,\theta)=P(\rho)\varphi(\theta)$

$$\therefore \frac{\rho^2P''+\rho P'}{P}=\frac{\varphi''}{-\varphi}=\lambda=\beta^2$$

$$\therefore\begin{cases}\varphi''(\theta)+\lambda\varphi(\theta)=0\\ \varphi(0)=\varphi(\pi)=0\end{cases}\qquad \therefore \varphi(\theta)=C_1\cos\beta\theta+C_2\sin\beta\theta$$

教学班号 05111003 姓名 周嫱 学号 1120101209.

$\therefore\ \varphi(0)=C_1=0.$

$\varphi(\pi)=C_2\sin\beta\pi=0.\quad\therefore\beta\pi=n\pi,\quad\therefore\beta=n\ (n=1,2,3\cdots)$

$\therefore\ \rho^2P''+\rho P'-\lambda P=0$

方程的通解为 $P(\rho)=D_n\rho^n+C_n\rho^{-n}.$

由 $\lim\limits_{\rho\to0}u=0$，~~$C_n=0.\ D_n=0$~~ $C_n=0$

则. $P(\rho)=D_n\rho^{+n}.$

$\therefore\ u_n(\rho,\theta)=A_n\rho^{+n}\sin n\theta$

$\therefore\ u(\rho,\theta)=\sum\limits_{n=1}^{\infty}A_n\rho^{+n}\sin n\theta$

又$\because$由边界条件 $u(a,\theta)=\theta(\pi-\theta)=\sum\limits_{n=1}^{\infty}A_na^{+n}\sin n\theta$

$\therefore\ A_n=\dfrac{2}{a^{+n}\pi}\int_0^{\pi}\theta(\pi-\theta)\sin n\theta\,d\theta$

$=\dfrac{2}{a^{+n}\pi}\left[\int_0^{\pi}\theta\pi\sin n\theta\,d\theta-\int_0^{\pi}\theta^2\sin n\theta\,d\theta\right]$

$\int_0^{\pi}\theta\pi\sin n\theta\,d\theta=-\dfrac{\pi\theta}{n}\cos n\theta\Big|_0^{\pi}+\int_0^{\pi}\dfrac{\pi}{n}\cos n\theta\,d\theta$

$=\dfrac{-\pi^2}{n}(-1)^n$

$\int_0^{\pi}\theta^2\sin n\theta\,d\theta=-\dfrac{\theta^2}{n}\cos n\theta\Big|_0^{\pi}+\int_0^{\pi}\dfrac{2\theta}{n}\cos n\theta\,d\theta$

$=-\dfrac{\pi^2}{n}(-1)^n+\dfrac{2\theta}{n^2}\sin n\theta\Big|_0^{\pi}+\dfrac{2}{n^3}\cos n\theta\Big|_0^{\pi}.$

$=-\dfrac{\pi^2}{n}(-1)^n+\dfrac{2}{n^3}\left[(-1)^n-1\right]$

$\therefore\ A_n=\dfrac{2}{a^{+n}\pi}\left\{-\dfrac{\pi^2}{n}(-1)^n+\dfrac{\pi^2}{n}(-1)^n-\dfrac{2}{n^3}\left[(-1)^n-1\right]\right\}$

$=\dfrac{-2}{a^{+n}\pi}\cdot\dfrac{2}{n^3}\left[(-1)^n-1\right]=\dfrac{-4}{n^3a^{+n}\pi}\left[(-1)^n-1\right]$

$\therefore\ u(\rho,\theta)=\sum\limits_{n=1}^{\infty}A_n\rho^{+n}\sin n\theta=\sum\limits_{n=1}^{\infty}\dfrac{4}{n^3\pi a^n}\left[1-(-1)^n\right]\rho^n\sin n\theta,\ (n=1,2,3\cdots)$